

10 POWER SUPPLY CONSIDERATIONS

With today's power-hungry processors and system peripherals, a good power source is critical for a smooth and reliable operation

1 Make sure you have enough juice

Very often, the standard power supplies that come with the system casings have a bare minimum power rating of 235 watts. While this is suited to very basic systems, the power requirements in your super system could soon supersede this low-end power supply. It is therefore wise to choose the highest rating your budget will allow—at least 300 watts, or in the case of peripheral-packed gaming rigs and workstations, even 500 watts.

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2 Trust the brand

There are numerous brands of power supplies on the market, many of which use very low-quality parts. Stay away from these and spend a little more on a reputable and reliable brand such as Codegen or Antec.

3 Watch the weight

As a rough rule of thumb, take two power supplies of the same rating but different brands and see which one is heavier. Usually, a heavier power supply implies that higher quality components such as heat sinks, capacitors and transformers are used and this translates into higher quality. Lighter power supplies usually comprise of

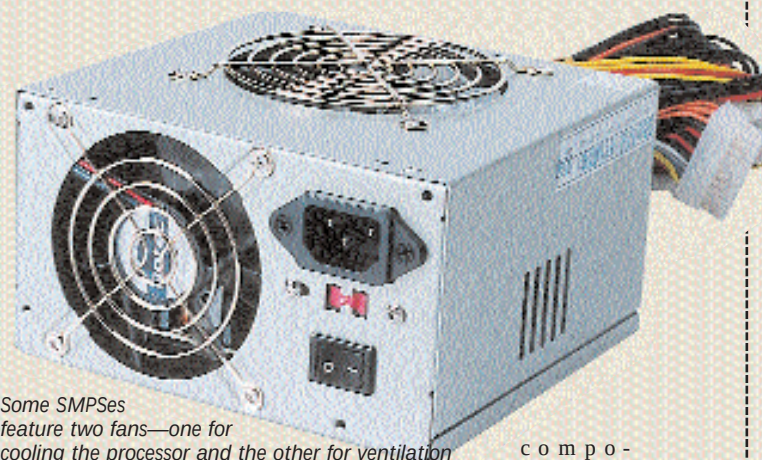
lower-quality components fabricated of cheaper and hence lighter materials. The lighter weight may also mean that some components, such as the heatsink, are completely left out.

4 Special processor compatibility

Depending upon your processor, make sure that your power supply can provide the juice required to run the processor. Look for any indication that the power supply is rated for use on AMD or Intel-based systems specifically. In the case of Pentium 4-based systems, look out for the presence of the ATX12V four-pin connector that is needed to power Pentium 4 processor-based computers.

5 Beware the bundles

Most power supplies bundled with the system casing are of low quality since their cost has to be integrated into the cost of the system. Beware of these power supplies and make sure that their rating is high enough for your particular processor, peripherals and internal system components. If need be, opt to trade in the integrated power supply with another of a higher rating, quality and brand, and offer to pay the price difference.



Some SMPSes feature two fans—one for cooling the processor and the other for ventilation

6 Go in for the highest

Even though you might not need the additional power, it makes sense to invest in a power supply that is rated slightly higher than for your application. Also, a higher rating does not mean that it necessarily draws more power—it will only draw as much power as is needed by your particular mix of system components.

7 How's the head count?

You may find out, much to your dismay, that the number of power supply heads from your SMPS may be lesser than the number needed by your hard disk, CD/DVD/CD-RW drive, etc. This is another factor you need to consider—your power supply should have enough power connectors, so that you do not need to invest in additional Y-cables for powering all your components.

8 SMPS cooler

In most systems, the SMPS fan shoulders the burden of cooling all the

components. This

makes it important for the fan to be powerful enough to handle the airflow in the system, without letting too much heat to be developed within the system.

9 Lower the noise

The SMPS fan tends to make a lot of noise due to the vibration. To prevent this, the SMPS unit must be firmly fastened to the casing, with no play in it. A loose attachment causes the power supply to vibrate excessively and can also cause component stability problems in the future.

10 Clean the fans regularly

The SMPS fan is one of the biggest components in the system and sometimes serves as the only duct for directing the heated air outside the system. Hence, it is very prone to dust and needs to be cleaned regularly to prevent dust build-up within the SMPS and the fan. A blower or brush does a competent job of dislodging any dirt or dust stuck on the fan blades.



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